Aegis Perception
Rain restoration experiment

RealRain-1k-H | Restormer fine-tuning | Seed 42
Report prepared: 7 October 2026

A completed initial restoration baseline: 20 epochs of supervised fine-tuning, validation-selected checkpoint, evaluation on all 224 test pairs, an untouched pretrained comparison, and three fixed qualitative examples. This report records the completed run; it is not an all-weather or deployment result.

Main result: selected checkpoint at epoch 15

Test condition (same 224 pairs)	PSNR (dB)	SSIM
Raw rainy input	22.2721	0.741376
Untouched pretrained Restormer	23.1265	0.794034
Fine-tuned Restormer (best.pt)	37.2565	0.972978

Fine-tuning improved mean test PSNR by **14.13 dB** over pretrained weights and **14.98 dB** over raw input. Mean SSIM improved by 0.1789 over pretrained weights.

Which trained file should be used?

best.pt: epoch 15, chosen by the highest validation PSNR. Use this checkpoint for the reported results, restoration inference, and later export experiments.
last.pt: epoch 20, the latest completed training state. Use it when resuming training. It is not the best checkpoint in this run.

Both files are in:

`artifacts/restoration/rain_restormer_seed42/`

Training job 47223 completed with exit code 0:0 and a reported SLURM elapsed time of 39 minutes 49 seconds. Fine-tuned test evaluation was job 47224; pretrained test evaluation was job 47225. Queue waiting time is excluded from runtime.

Evidence: run.json, history.json, metrics_test.json, metrics_test_pretrained.json, checkpoint contents, and archived logs. Job completion/elapsed time is also supported by the user-supplied sacct output.

Data, architecture, and training recipe

Item	Recorded value
Dataset and split	RealRain-1k-H: 784 train / 112 validation / 224 test pairs
Manifest	data/manifests/realrain1k.csv
Local data root	data/raw/rain/realrain1k/RealRain-1k-H/
Model	Restormer; 26,126,644 trainable parameters
Architecture	Base width 48; blocks [4,6,6,8]; heads [1,2,4,8]; 4 refinement blocks; expansion 2.66; WithBias LayerNorm; convolution bias disabled
Initialization	Official deraining.pth weights loaded strictly; all model parameters fine-tuned
Training images	Synchronized 128 x 128 random crops; paired horizontal flips; batch 4; 4 data workers
Schedule	20 epochs; 196 batches per epoch; 3,920 optimizer updates; seed 42
Objective	L1 reconstruction, weight 1.0; no discriminator or adversarial term
Optimizer	AdamW; learning rate 0.0001; betas (0.9,0.999); weight decay 0.0001
Learning rate and gradients	Cosine schedule toward 0.000001; gradient clipping 1.0
Precision	BF16 autocast for CUDA model execution; reconstruction loss computed from float predictions

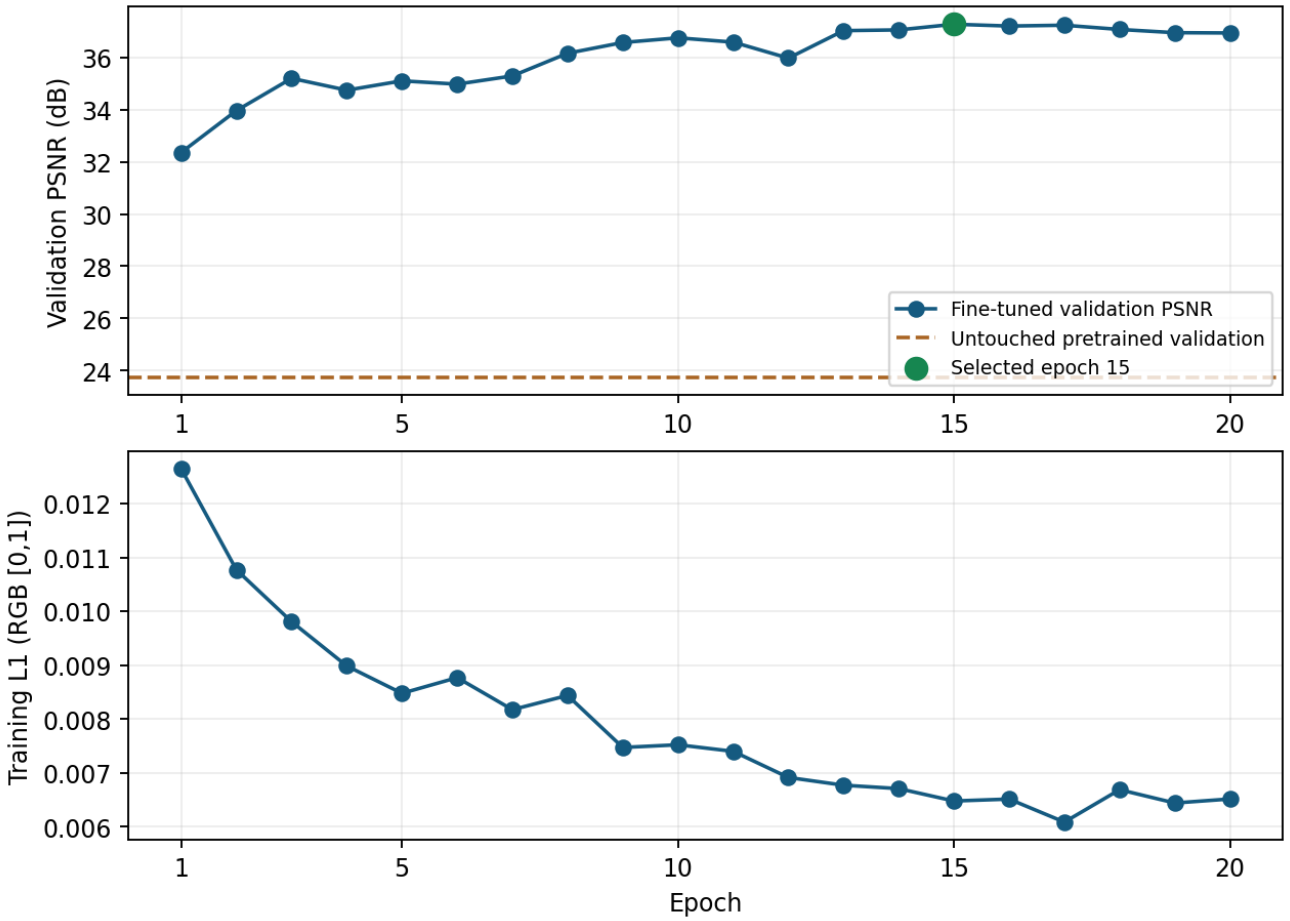
Project tensors use [-1,1]. For Restormer, the trainer multiplies their mean absolute reconstruction error by 0.5, making the training objective equivalent to mean L1 on RGB [0,1]. The logged validation L1 uses [-1,1] without this factor, so its magnitude should not be compared directly to the logged training L1.

Evaluation protocol

Validation and test use native image resolution, batch size 1, and BF16 model execution on CUDA. Input padding meets the architecture dimensions and is removed before scoring. Outputs are converted to RGB [0,1] and clamped. PSNR uses per-image RGB MSE with a 1e-12 floor, followed by an arithmetic mean across images; SSIM uses scikit-image with channel_axis=-1 and data_range=1. These are RGB scores, not Y-channel scores.

All 112 validation images were checked before optimization. Checkpoint selection used validation only. The pretrained and fine-tuned test files have matching manifests and evaluation protocols. Numeric metrics are calculated before saving the 8-bit PNG illustrations; figure export is not the scoring path.

Learning curve and resource usage

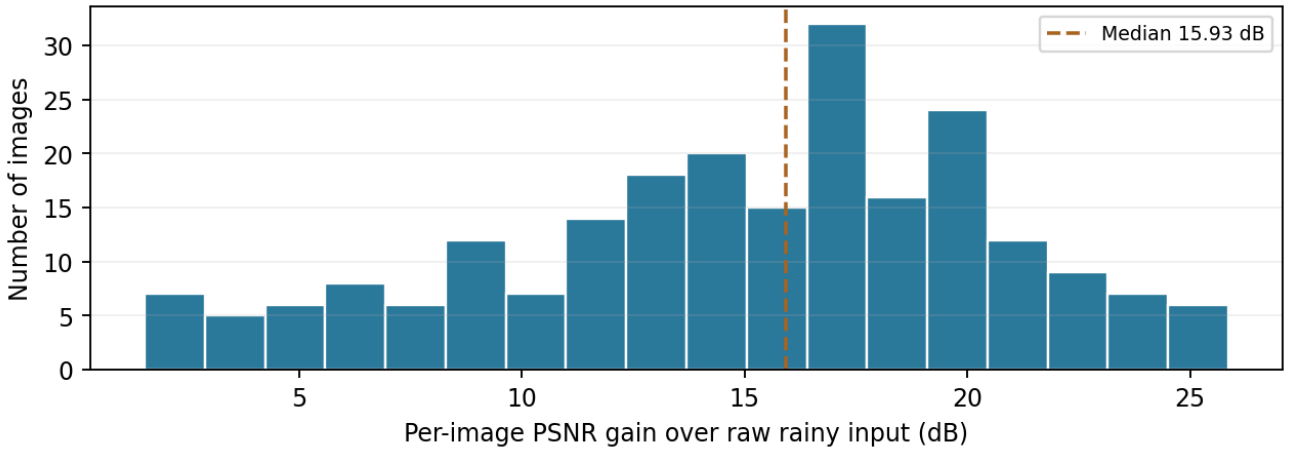


Measurement	Value
Pretrained validation PSNR	23.7324 dB
Validation: epoch 1 / best epoch 15 / epoch 20	32.3582 / 37.2741 / 36.9392 dB
Hardware and environment	NVIDIA A100-SXM4-40GB; Python 3.10.18; Torch 2.5.1+cu118; CUDA build 11.8
SLURM resource request	1 full A100 40 GB; 4 CPU cores; 30000M system RAM; longq
Mean training / total epoch time	40.9 s / 114.5 s
Peak allocated CUDA memory (logged)	22.28 GiB; not total process/reserved GPU memory

Epochs 15-20 broadly plateaued; epoch 20 was 0.335 dB below the selected checkpoint. This supports keeping epoch 15 for this run. It does not prove that a longer or different schedule cannot improve performance, or establish overfitting from this curve alone.

Per-image behavior and limits of the average

Test statistic	Value
PSNR improved over raw input	224 / 224 images
Median / 10th-percentile PSNR gain over raw	15.93 / 6.15 dB
Minimum / maximum final PSNR	18.19 / 49.68 dB
SSIM below raw input	2 / 224: test/107, test/863
PSNR below pretrained output	2 / 224 images



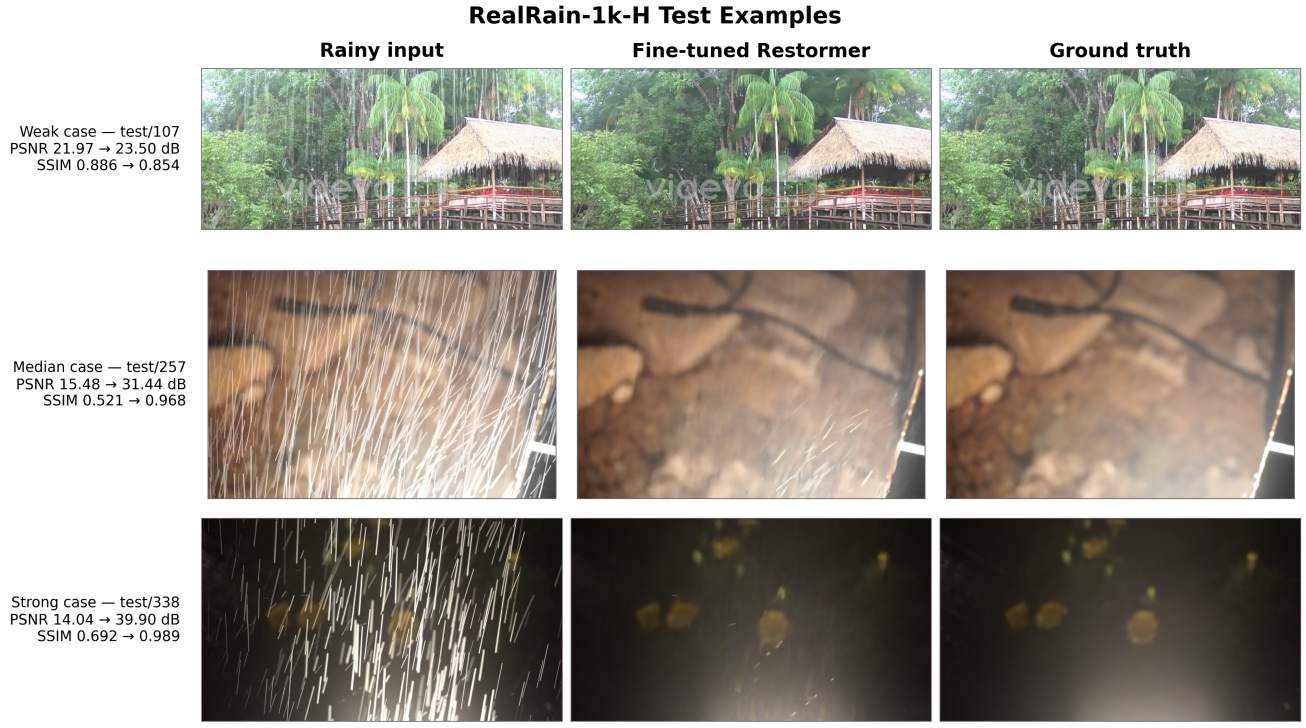
Example	Raw PSNR	Final PSNR	Raw SSIM	Final SSIM
test/52	15.92	18.19	0.7708	0.8324
test/45	19.29	21.57	0.8194	0.9524
test/863	19.19	21.76	0.8412	0.8334
test/107	21.97	23.50	0.8858	0.8541

Observed: most pairs improve substantially, but a small number remain difficult. In test/107, PSNR rises while SSIM falls, so a single metric does not describe every change in structure or appearance. The illustrated result also changes brightness/contrast; possible overcorrection is a hypothesis to investigate, not a verified failure mechanism.

Interpretation: fine-tuning adapted pretrained Restormer effectively to this dataset. The improvement does not isolate the effect of L1 versus other losses; there was no loss ablation. BF16 completed successfully, but no matched FP32 comparison was run.

Fixed qualitative comparison

Columns: rainy input, fine-tuned Restormer, paired ground truth. The three examples were selected from recorded test results by PSNR gain: minimum (test/107), upper-middle sorted entry (test/257), and maximum (test/338). They are not three randomly sampled examples, and the weak case means weakest gain rather than lowest absolute PSNR.



Selected test case	Gain over raw	Interpretation
test/107	1.53 dB	Weakest PSNR gain; SSIM decreased
test/257	15.95 dB	Near the median gain; visible residual streaks
test/338	25.86 dB	Largest PSNR gain; substantial streak removal

The original restored PNGs retain their native sizes. The report figure reduces them for display only; quantitative evaluation was performed at native resolution. Visual agreement on these examples is supporting evidence, not a substitute for the full test-set metrics.

Full-size comparison files:
`qualitative/rain_restoration_comparison.png`
`qualitative/rain_restoration_comparison.pdf`

Checkpoints, reproducibility, and scope

File	Epoch	Use
best.pt	15	Recommended for inference; highest validation PSNR
last.pt	20	Latest state; use to resume training
epoch_005.pt / epoch_010.pt	5 / 10	Intermediate training snapshots
epoch_015.pt	15	Periodic snapshot of the selected epoch
epoch_020.pt	20	Periodic snapshot of the final epoch
pretrained.pt	0	Untouched initialization packaged for project evaluation

Trained checkpoints are approximately 300 MiB each because they contain generator weights, AdamW optimizer state, scheduler/scaler state, RNG state, configuration, history, and validation metadata. The pretrained checkpoint is approximately 100 MiB and has no optimizer state. These are different training states, not multiple architectures. No checkpoint was deleted or renamed for this report.

Laptop artifact directory:
`/home/bms/Desktop/BTP_Aegis-Perception/artifacts/restoration/rain_restormer_seed42/`
DGX artifact directory:
`/scratch/cs23b051/aegis-perception/artifacts/restoration/rain_restormer_seed42/`

Core evidence: **run.json** (recipe/environment/checksums), **history.json** (20 epochs), **baseline_validation.json**, **metrics_test.json**, **metrics_test_pretrained.json** (224 per-image records each), **environment.txt**, **source_code.tar.gz**, training/evaluation logs, and restored images.

Recorded Git commit:
`3aa4ef4e5c6bb75e675293791b65cd89c8eb8948`

Training source SHA-256:
`4314bcb8b0fd729bee1017ecbc730662a9195a57d74f5d00ad4d69c70c9fbdb3`

Manifest SHA-256:
`b03a7799863f5e50e79140940fcb329cfbe368267a76a62e5a6e61c8869f31c5`

Official pretrained weight SHA-256:
`225ed9acf80b3527d7080e5182918f642a23a78ef8593ad700b7e924bd660681`

The run includes local source changes beyond the recorded Git commit; the recorded source checksum and retained code archive are needed alongside the commit. BF16 reduced inference memory; large depthwise convolutions were split by independent channels to avoid the older CUDA kernel indexing limit while retaining spatial context.

Supported conclusion: a successful single-weather, single-seed Restormer fine-tuning baseline on the recorded RealRain-1k-H split. Cross-split near-duplicate scene auditing, seeds 43/44, unseen-dataset evaluation, multi-weather training, detector benefit, FP32/BF16 comparison, and quantized hardware measurements remain unverified. Single-seed results have no across-seed uncertainty estimate; they are not evidence of state-of-the-art performance or submission readiness.